



PharmDB

Pharmacy Database System

CIS 421

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Source Code: <https://github.com/ryanaiien/cis-421-project>

Application Background

- A regional group of retail pharmacies operating across Michigan
- Key actors: pharmacies, employees, doctors, patients, drugs, and drug manufacturers
- Core operations: prescription fulfillment, inventory management, staffing, and manufacturer contracts
- Each pharmacy manages its own inventory, pricing, and manufacturer contracts.

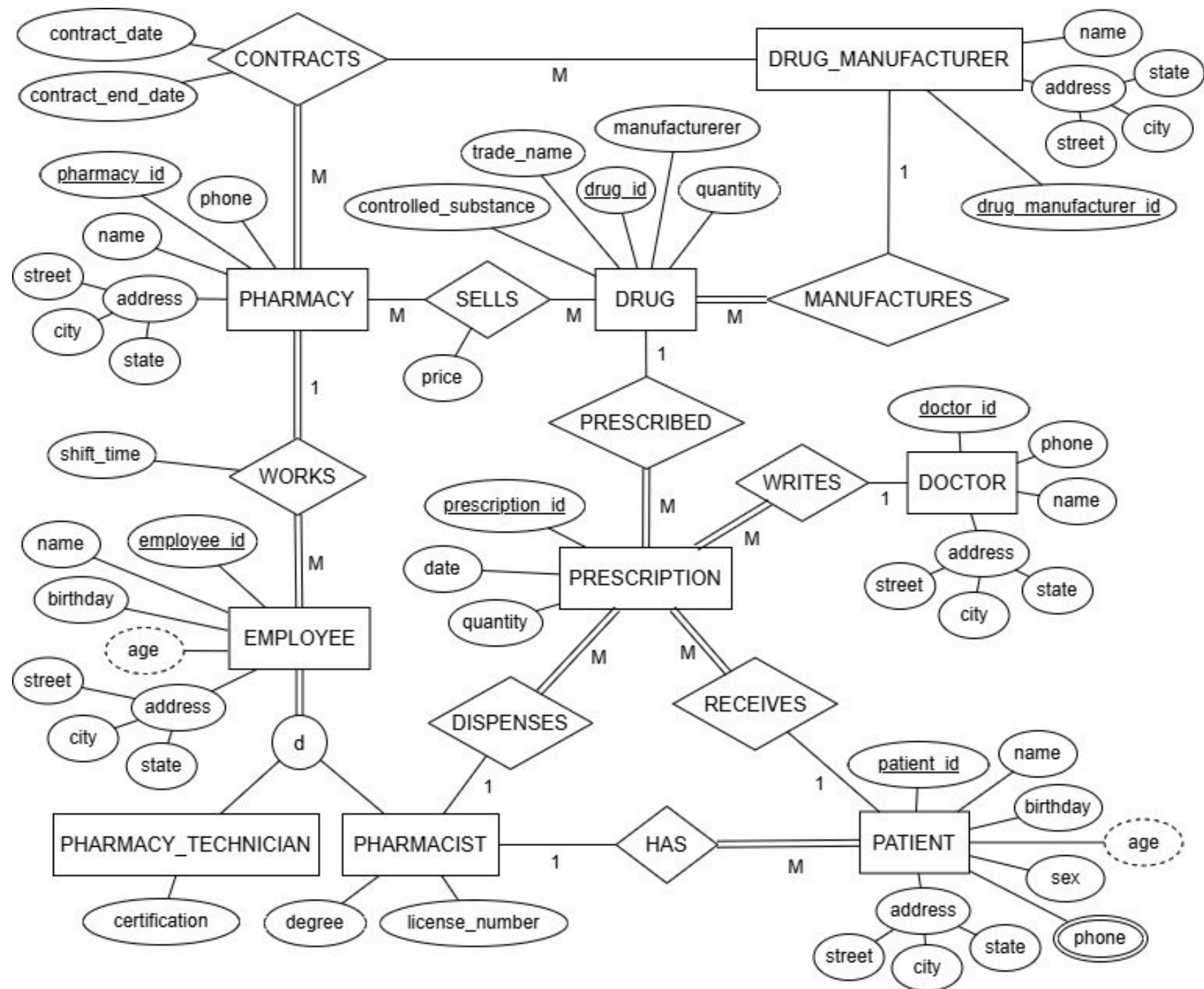


Database Requirements

- Track each pharmacy's location, staff, inventory, and pricing
- Track employees using an IS-A hierarchy: pharmacist or pharmacy technician
- Track doctors who write prescriptions and patients who receive them
- Track patient phone numbers as a multi-valued attribute
- Track drugs, drug manufacturers, and pharmacy-manufacturer contracts
- Track prescriptions linking doctor, patient, pharmacist, and drug



Entity Relationship Diagram



Schema

- Pharmacy(pharmacy_id, name, street, city, state, phone)
- Employee(employee_id, pharmacy_id, name, street, city, state, birthday, employee_type, shift_time)
- Pharmacist(employee_id, degree, license_number)
- PharmacyTechnician(employee_id, certification)
- Doctor(doctor_id, name, specialty, phone, street, city, state)
- Patient(patient_id, name, sex, insurance, birthday, street, city, state, primary_pharmacist_id)
- PatientPhone(patient_id, phone_number)
- Drug(drug_id, trade_name, manufacturer_id, quantity, controlled_substance)
- DrugManufacturer(manufacturer_id, name, street, city, state)
- PharmacySells(pharmacy_id, drug_id, price)
- Prescription(prescription_id, doctor_id, pharmacist_id, patient_id, drug_id, prescription_date, quantity)
- PharmacyContract(pharmacy_id, manufacturer_id, contract_date, contract_end_date)

Sample Database

Table	Rows
Pharmacy	5
Employee	15
Pharmacist	6
PharmacyTechnician	9
Doctor	8
Patient	20
PatientPhone	26
Drug	30
DrugManufacturer	8
Prescription	40
PharmacySells	50
PharmacyContract	19

- Data is Michigan-based: Dearborn, Detroit, Ann Arbor, Ypsilanti, Lansing
- Drug names are real FDA-approved drugs (Metformin, Lisinopril, Atorvastatin, etc.)
- The row counts reinforce two key modeling choices: 6 pharmacists and 9 technicians under Employee, plus 26 patient phone numbers for 20 patients
- 40 prescriptions distributed across all 5 pharmacies, enough to make aggregate queries meaningful without making the dataset hard to explain

Query 3: Pharmacies stocking all Pfizer drugs

Find the names of the pharmacies where there does NOT EXIST a Pfizer drug that the pharmacy does NOT sell.

```
SELECT Pharmacy.name FROM Pharmacy
WHERE NOT EXISTS (
    SELECT Drug.drug_id FROM Drug
    JOIN DrugManufacturer ON Drug.manufacturer_id =
DrugManufacturer.manufacturer_id
    WHERE DrugManufacturer.name = 'Pfizer'
    AND NOT EXISTS (
        SELECT PharmacySells.drug_id FROM PharmacySells
        WHERE PharmacySells.pharmacy_id = Pharmacy.pharmacy_id
        AND PharmacySells.drug_id = Drug.drug_id
    )
);
```

→ HealthPlus Pharmacy

Query 12: Bulk price update

Update drug prices by manufacturer

```
UPDATE PharmacySells
SET price = ROUND(price * 1.10, 2)
WHERE drug_id IN (
    SELECT drug_id FROM Drug
    JOIN DrugManufacturer ON Drug.manufacturer_id =
DrugManufacturer.manufacturer_id
    WHERE DrugManufacturer.name = 'AstraZeneca'
);
```

Result: Omeprazole at pharmacy 2 changed from 15.99 to 17.59

User Interface Demo

- A web-based Database Admin UI for PharmacyDB.db using Django
- Django connects directly to the same SQLite file, with no data duplication
- Minimal frontend code was written. Django generates the CRUD interface by inspecting the database and creating matching Python class models

PharmDB		
Database Administration		
AUTHENTICATION AND AUTHORIZATION		
Groups	+ Add	✎ Change
Users	+ Add	✎ Change
PHARMACY		
Doctors	+ Add	✎ Change
Drug Manufacturers	+ Add	✎ Change
Drugs	+ Add	✎ Change
Employees	+ Add	✎ Change
Patient Phones	+ Add	✎ Change
Patients	+ Add	✎ Change
Pharmacies	+ Add	✎ Change
Pharmacists	+ Add	✎ Change
Pharmacy Contracts	+ Add	✎ Change
Pharmacy Sells	+ Add	✎ Change
Pharmacy Technicians	+ Add	✎ Change
Prescriptions	+ Add	✎ Change

Django Admin Demo

PharmDB

WELCOME, ADMIN. [CHANGE PASSWORD](#) / [LOG OUT](#)

[Home](#) > [Pharmacy](#) > Patients

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION

Groups [+ Add](#)

Users [+ Add](#)

PHARMACY

Doctors [+ Add](#)

Drug Manufacturers [+ Add](#)

Drugs [+ Add](#)

Employees [+ Add](#)

Patient Phones [+ Add](#)

Patients [+ Add](#)

Pharmacies [+ Add](#)

Pharmacists [+ Add](#)

Pharmacy Contracts [+ Add](#)

Pharmacy Sells [+ Add](#)

Pharmacy Technicians [+ Add](#)

Prescriptions [+ Add](#)

Select patient to change

Search

Action:

Run

 0 of 20 selected

☐	NAME	SEX	INSURANCE	BIRTHDAY	AGE	CITY	STATE
☐	Paula West	F	Humana	Feb. 18, 1977	49	Detroit	MI
☐	Oscar Diaz	M	UnitedHealth	June 24, 1959	66	Dearborn	MI
☐	Nina Patel	F	Cigna	Jan. 13, 1992	34	Lansing	MI
☐	Marcus Young	M	Aetna	Sept. 30, 1968	57	Ypsilanti	MI
☐	Lana Morris	F	BlueCross	April 5, 1987	39	Ann Arbor	MI
☐	Kyle Adams	M	Humana	Nov. 21, 1975	50	Detroit	MI
☐	Jenny Liu	F	UnitedHealth	July 16, 1991	34	Dearborn	MI
☐	Ivan Cruz	M	Cigna	March 9, 1983	43	Lansing	MI
☐	Helen Park	F	Aetna	Oct. 14, 1995	30	Ypsilanti	MI
☐	George Bell	M	BlueCross	May 28, 1962	63	Ann Arbor	MI
☐	Fiona Grant	F	Humana	Aug. 7, 1970	55	Detroit	MI

ADD PATIENT +

FILTER

[Show counts](#)

↓ By sex

[All](#)

[F](#)

[M](#)

↓ By insurance

[All](#)

[Aetna](#)

[BlueCross](#)

[Cigna](#)

[Humana](#)

[UnitedHealth](#)

↓ By state

[All](#)

[MI](#)

Team Responsibilities

Member	Responsibilities
Ryan	Refined the entity design, expanded the sample dataset, developed queries 10-14, built the Django admin interface, and prepared the written report.
Gavin	Designed the ER diagram, finalized the relational schema and DDL, prepared the insert statements and database setup, developed queries 1-9, captured query outputs, and prepared the presentation materials.

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Thank You!